

INTRODUCTION

Animals can use elastic energy to move incredibly fast, circumventing limitations of their muscles.

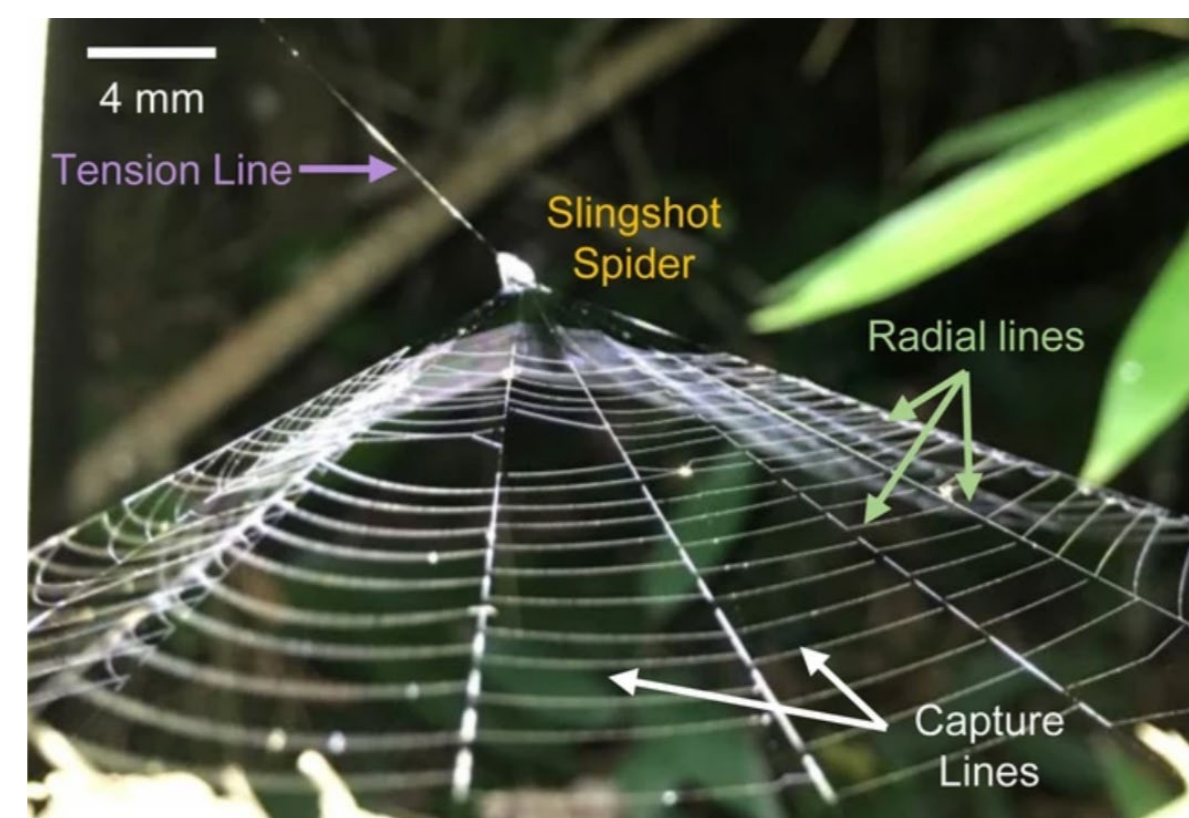
For example, slingshot spiders slowly load and quickly release their elastic webs, which recoil in 10 ms with a maximum acceleration of $> 10^3 \text{ m/s}^2$

Characterizing elastic material performance under the relevant loading/unloading conditions is important for understanding the biomechanics of these systems and designing bioinspired elastic mechanisms.

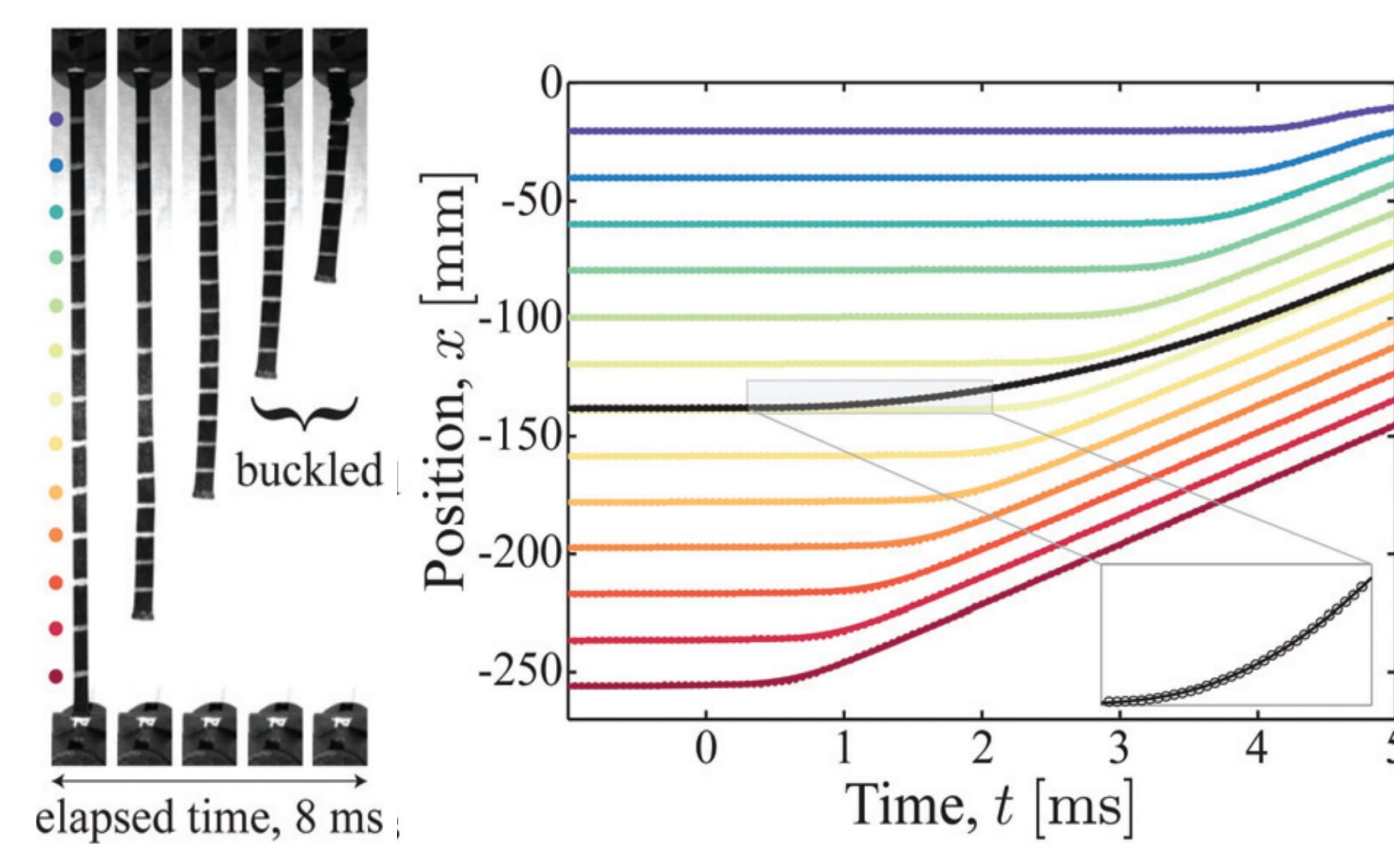
Recoil experiments have been used to measure the extreme limits of elastic performance.

Recoil experiments involve stretching and releasing a strip of material. Dots are also marked along the material to track how the movement is different at different points of the strip.

Previous work focused on the elastic response of the material, but the role of energy dissipation due to viscoelasticity at these extreme unloading conditions is not well understood.

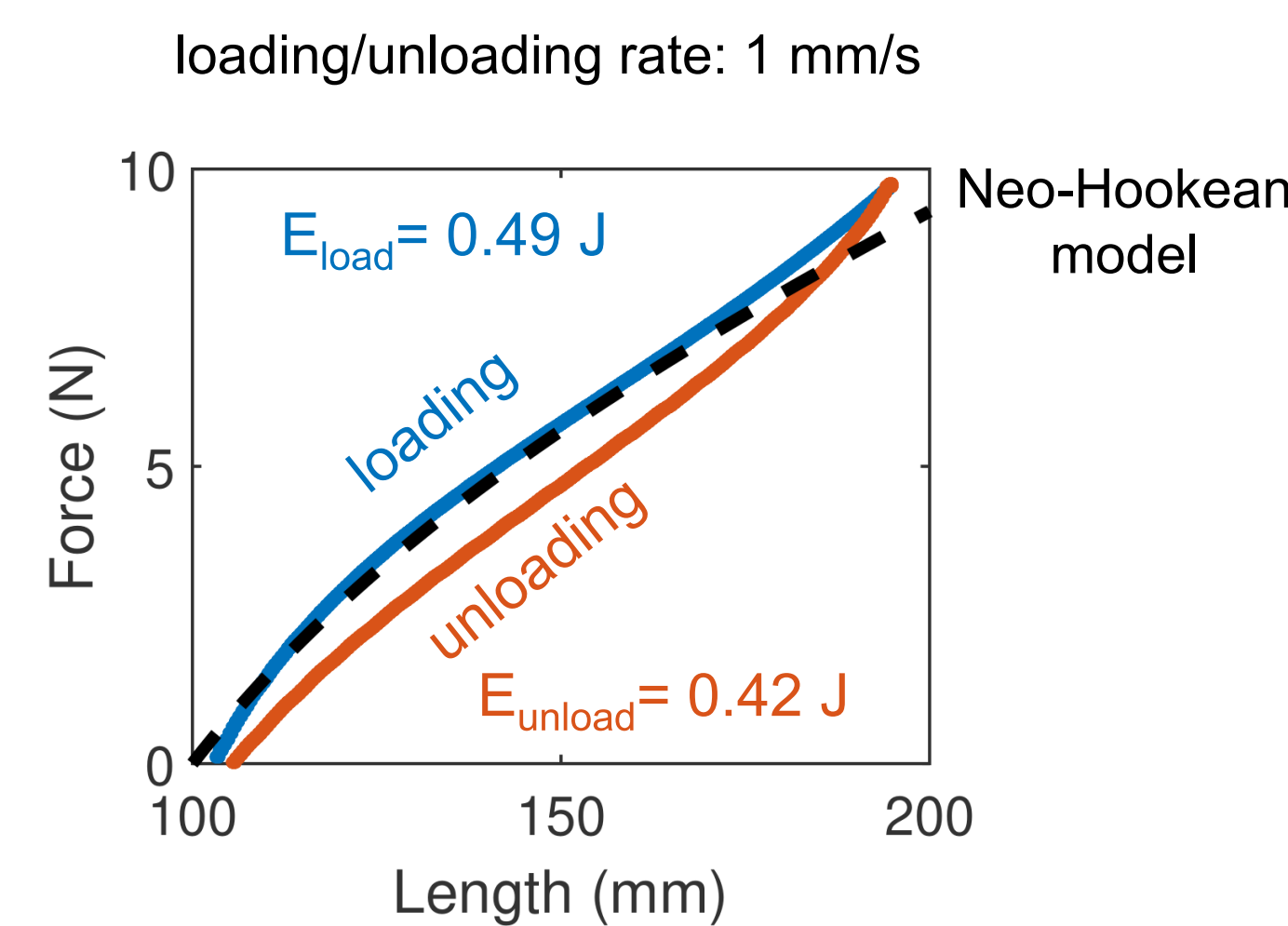
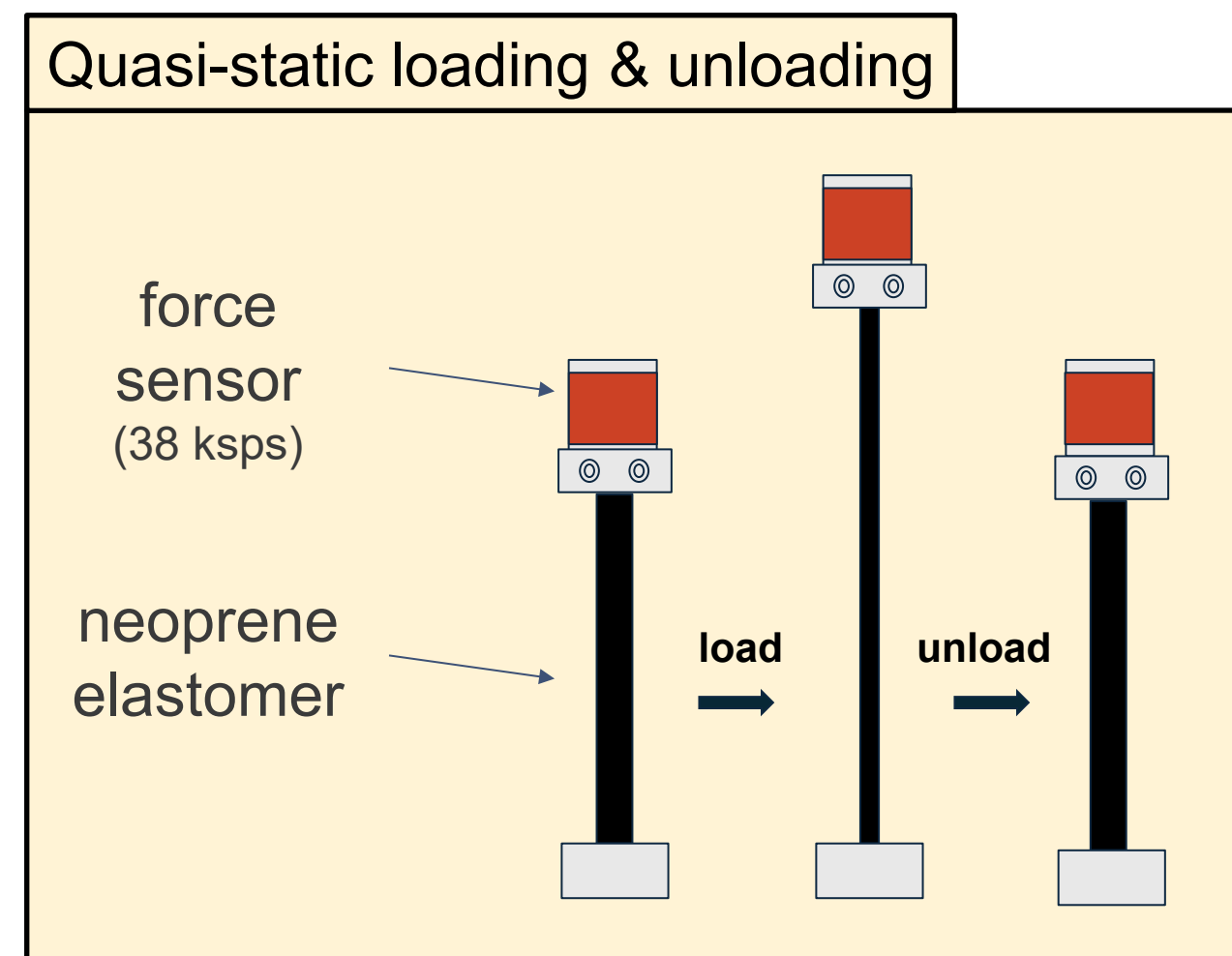


Challita et al. (2021)



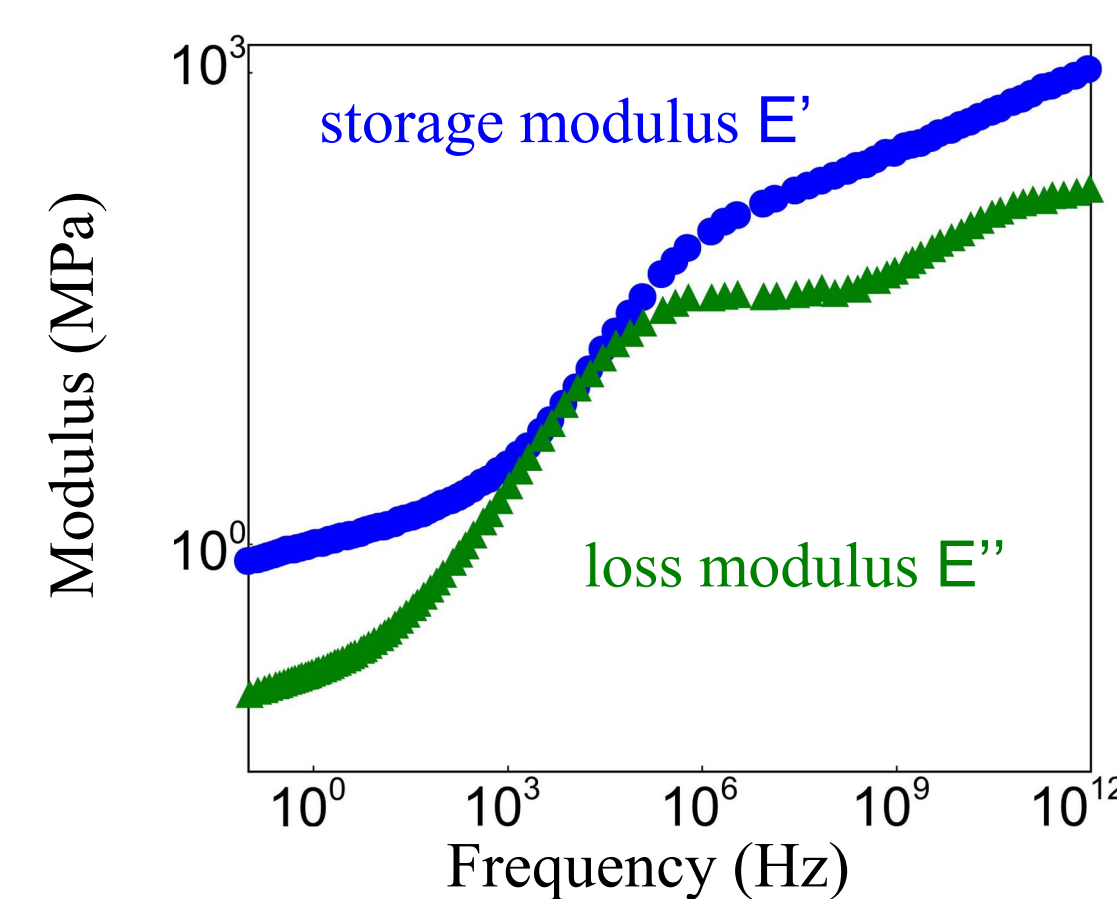
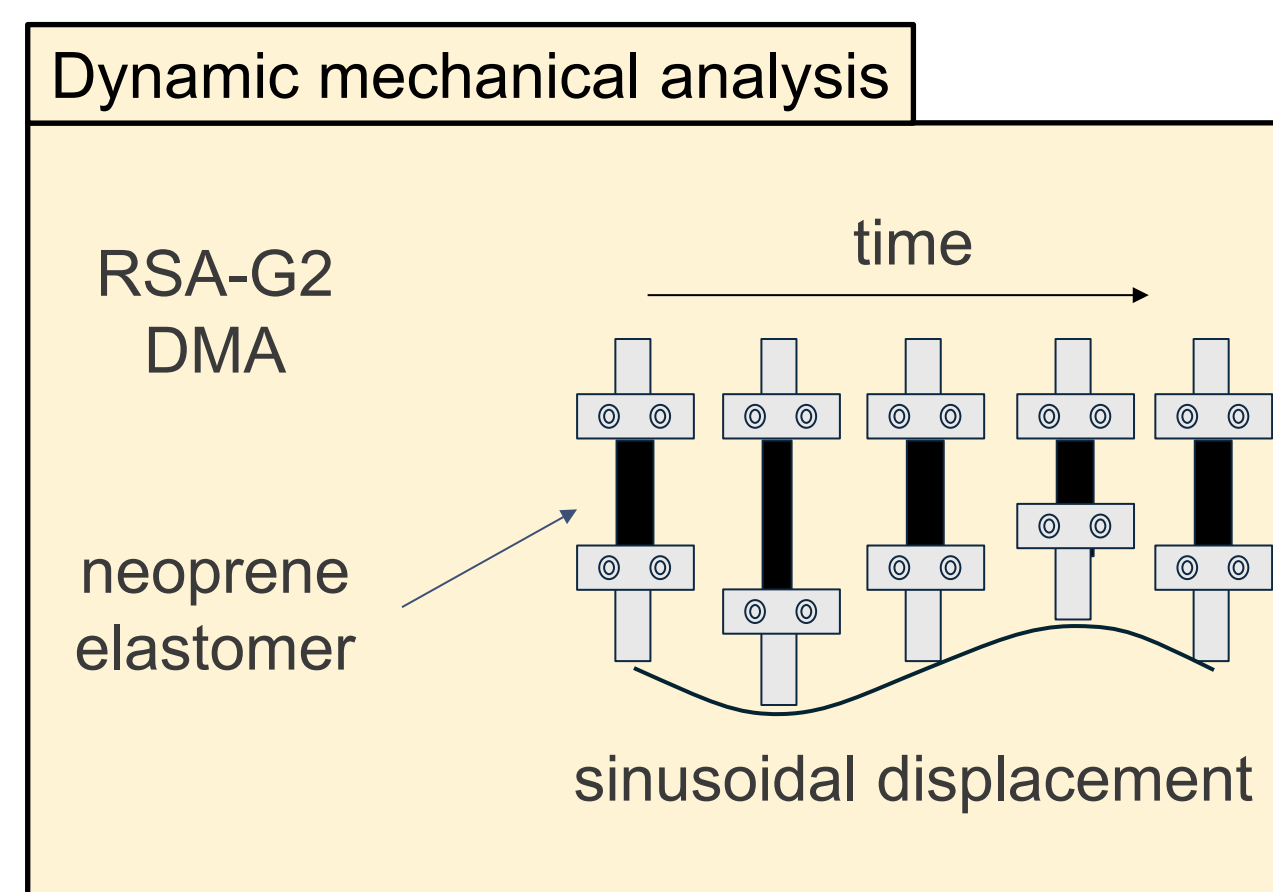
Ilton et al. (2019)

MATERIAL CHARACTERIZATION

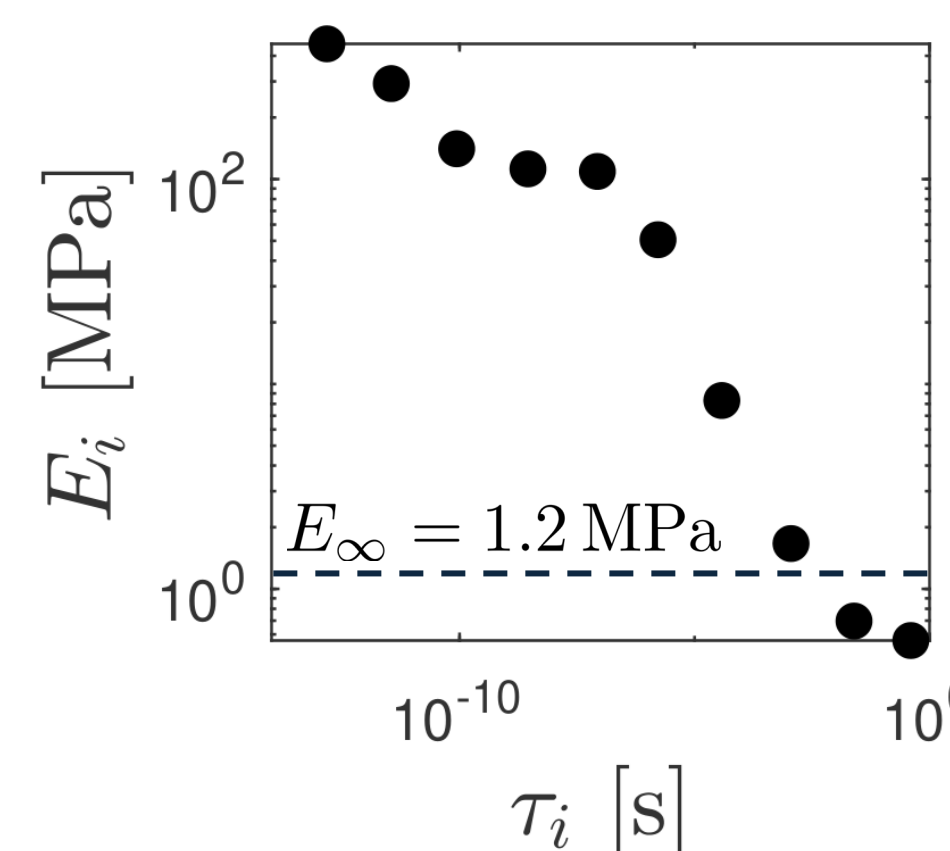
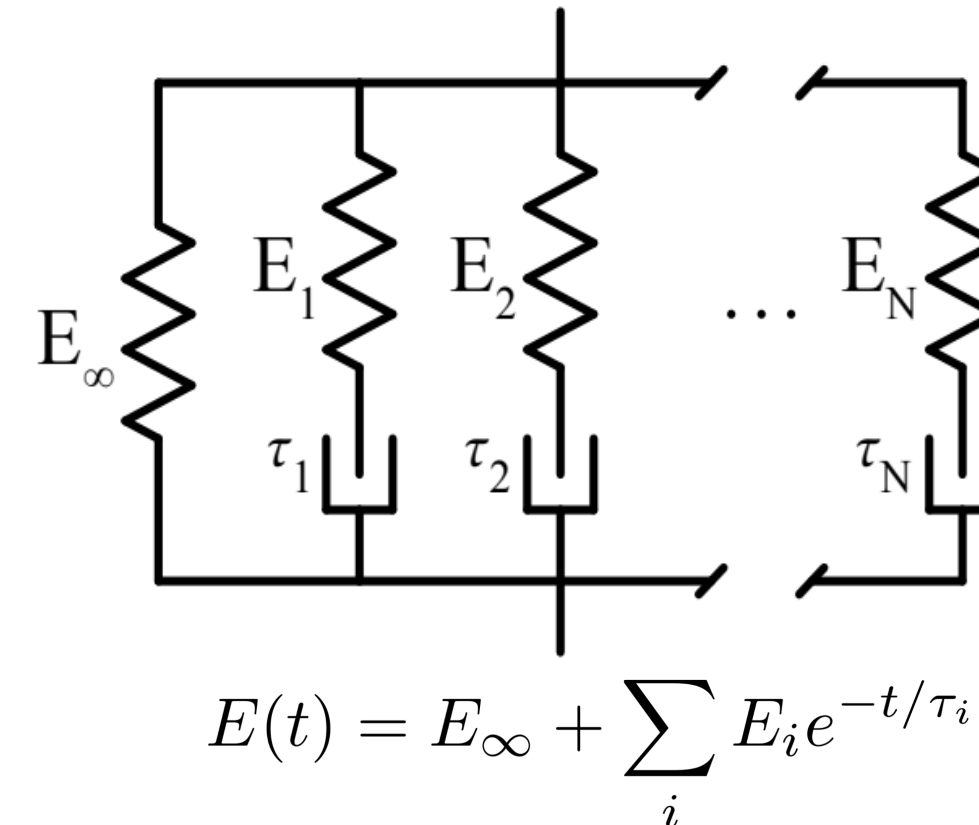


$$\text{resilience} = \frac{E_{\text{unload}}}{E_{\text{load}}} = 0.86$$

$$\sigma_{\text{elastic}}(\epsilon) = \frac{E_{\text{elastic}}}{3} \left(1 + \epsilon - \frac{1}{(1 + \epsilon)^2} \right)$$



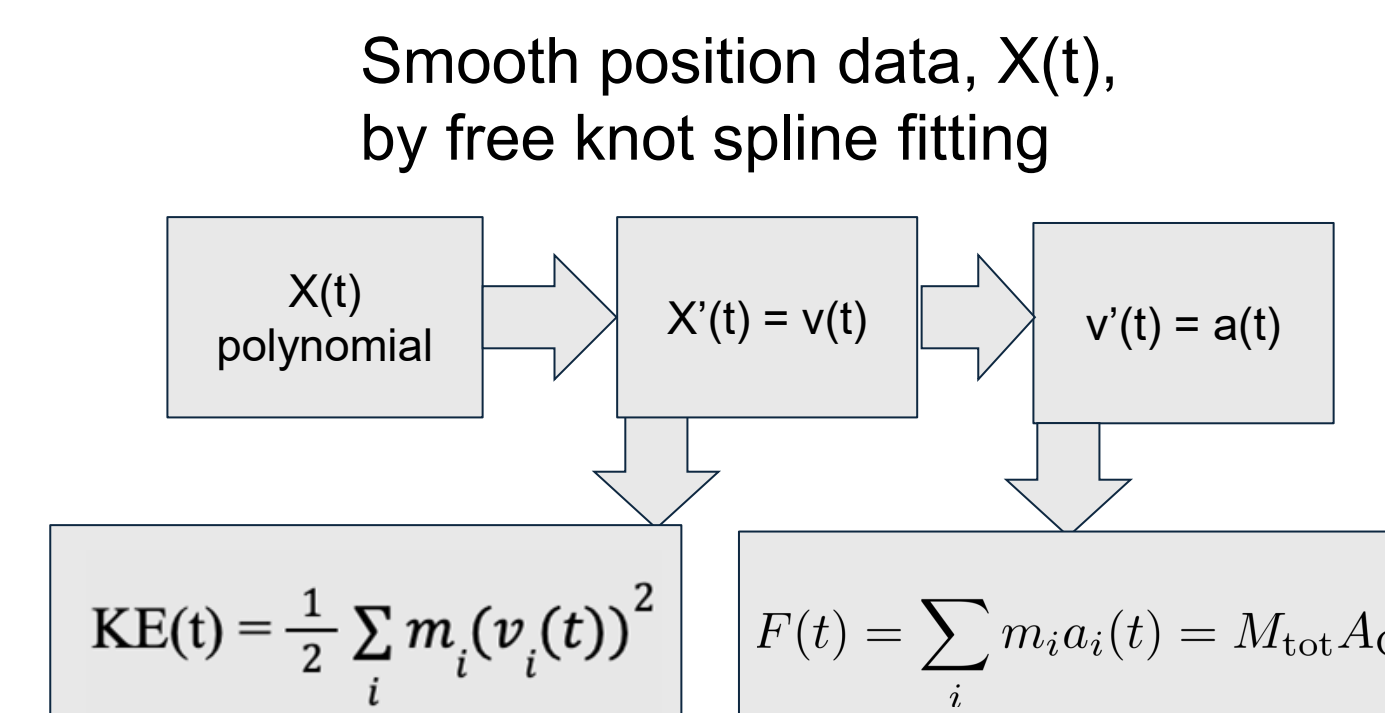
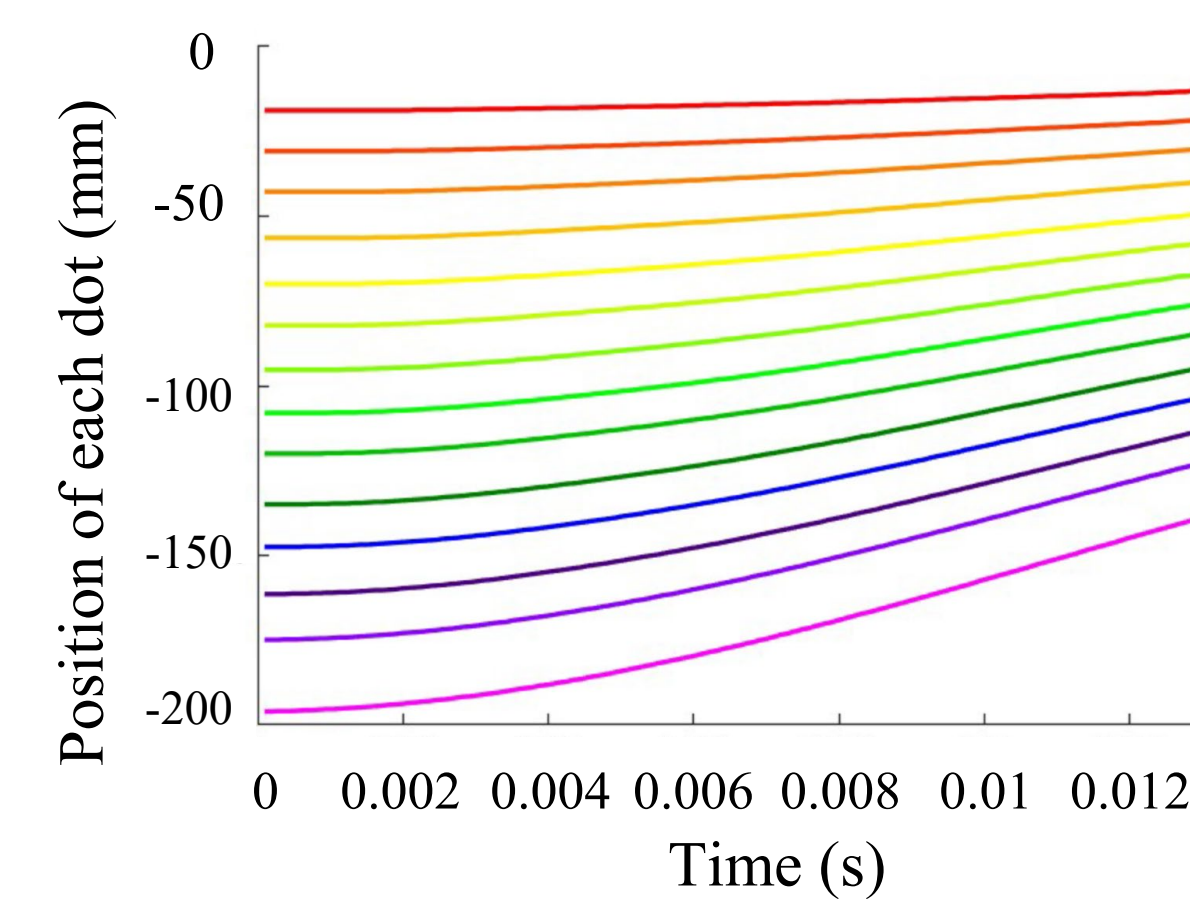
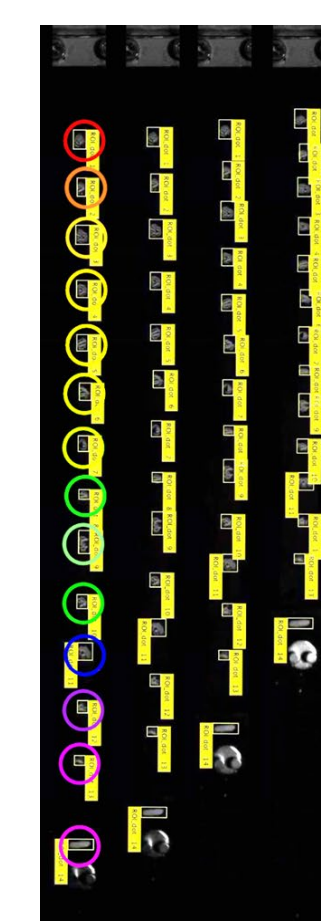
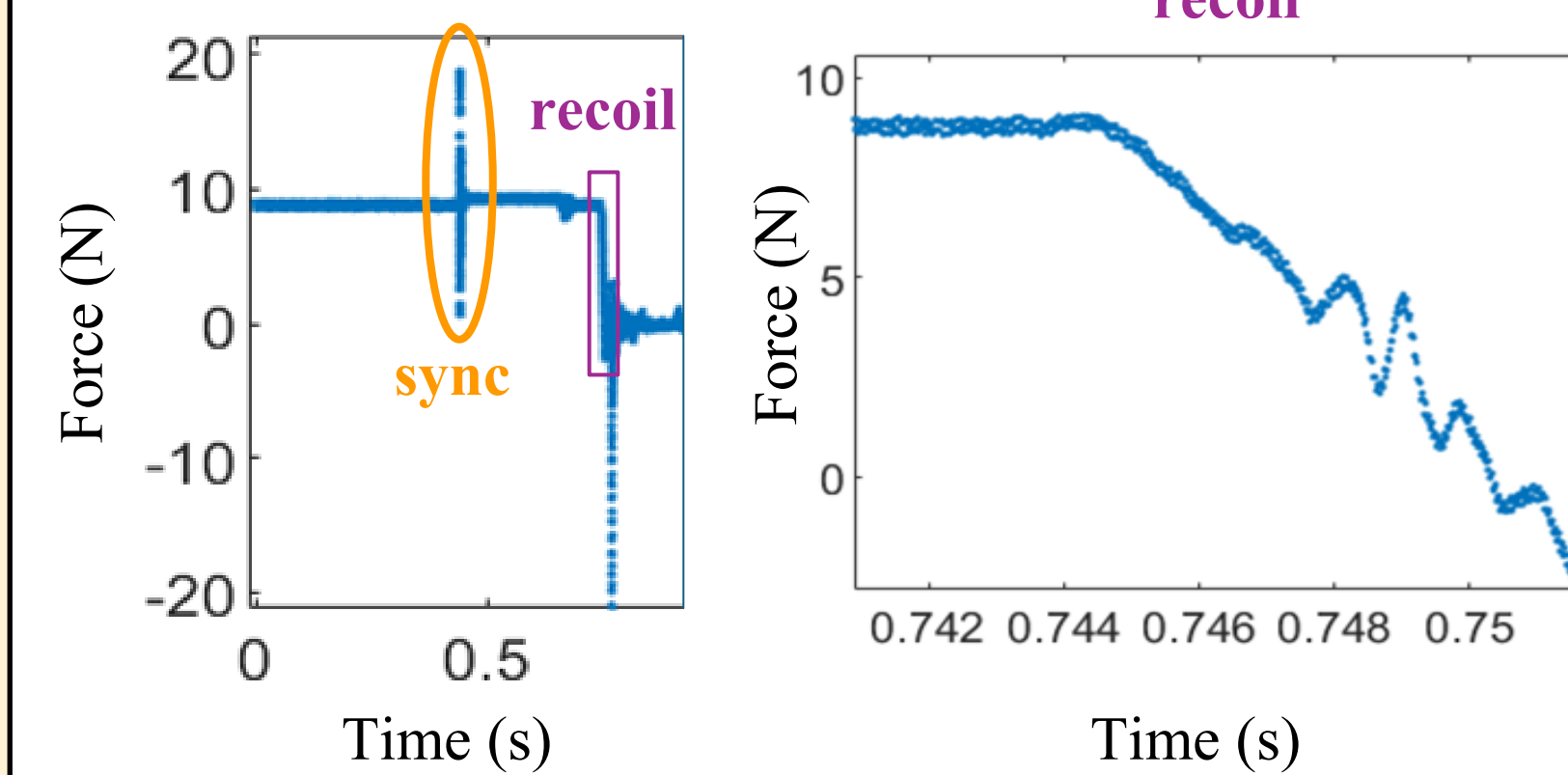
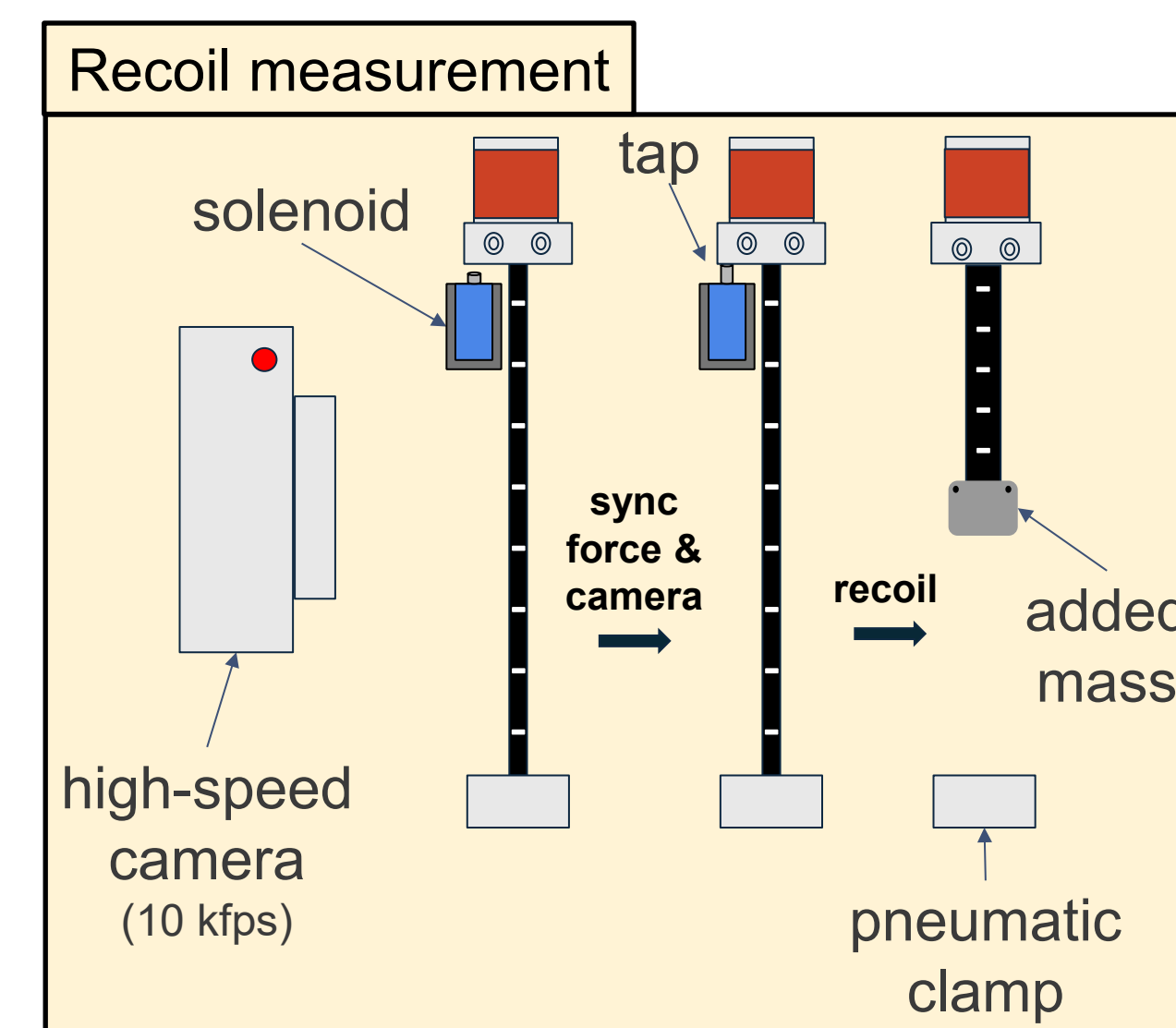
Generalized Maxwell Model



PROJECT AIM

In this work, we directly measure the high-rate large deformation unloading of elastic materials and compare to a simulation based on the independently-determined viscoelastic properties of the material.

DIRECT RECOIL MEASUREMENT

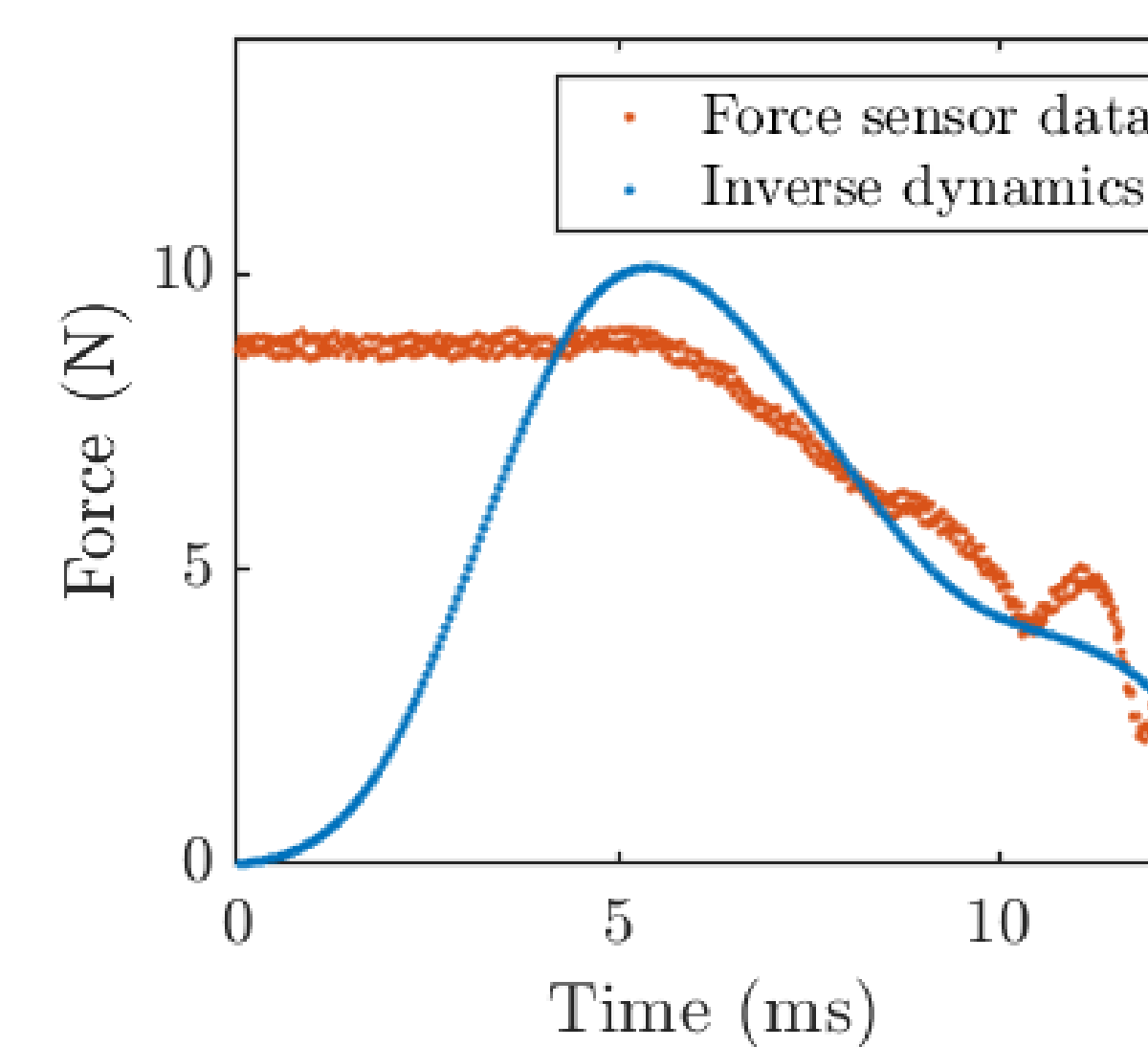


MEASURING FORCE AT HIGH-RATE

We measured force in two ways in our experiment:
(1) force sensor (direct measurement)
(2) inverse dynamics ($F = M_{\text{tot}} A_{\text{CM}}$)

Difference between the two forces for the first 5 ms suggests other forces acting on the sample, e.g. friction from the clamp during release and air drag.

Overshoot of inverse dynamics compared to force sensor indicates a potential systematic error.



RECOIL SIMULATION

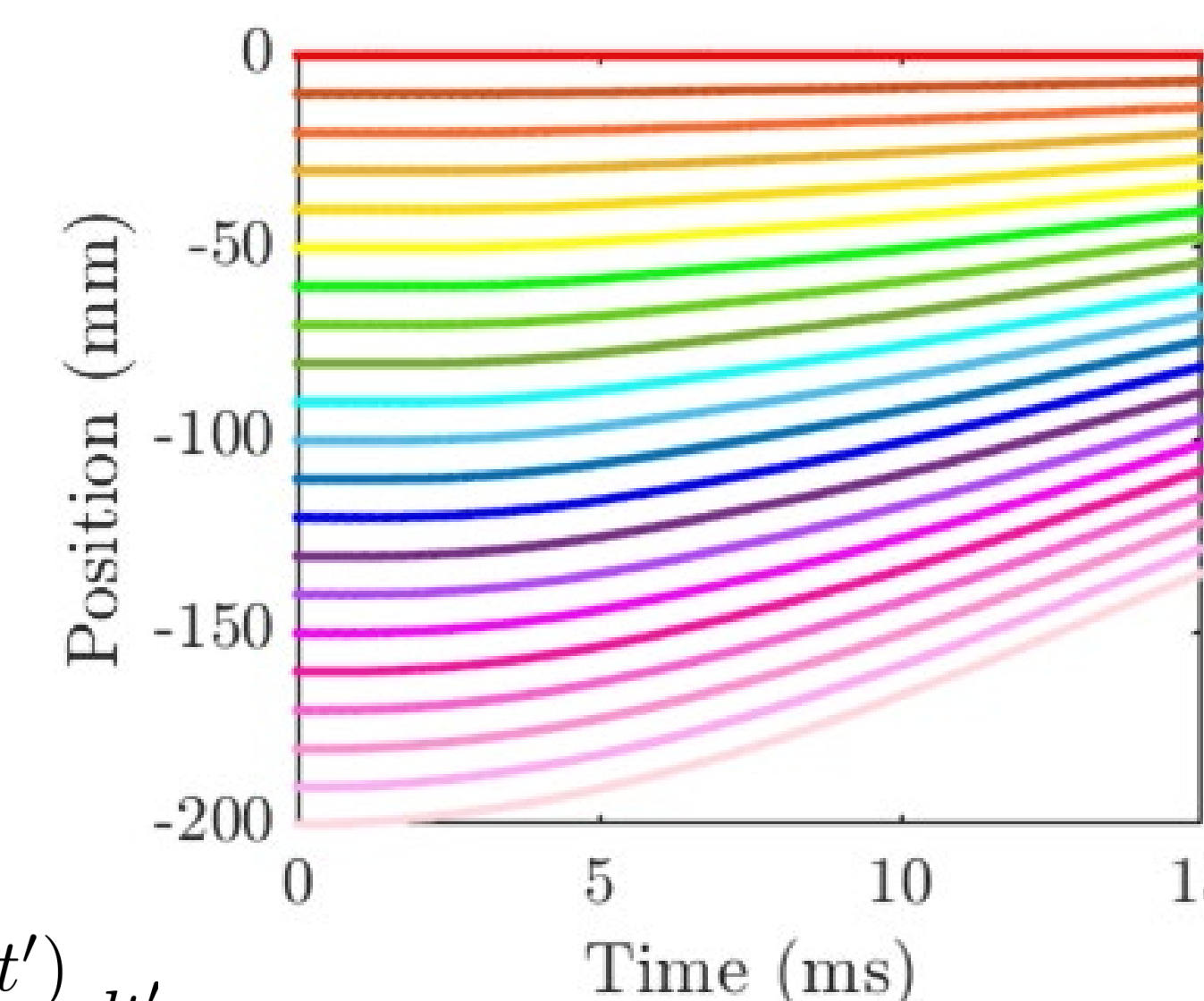
To simulate recoil, we discretize a 1D spatial grid and assign each material point the measured $E(t)$ and E_{elastic} . We then solve Newton's second law for the displacement U at each point:

$$\text{gradient of stress} \rightarrow \frac{\partial \sigma}{\partial x} = \rho \frac{\partial^2 U}{\partial t^2} \leftarrow \text{acceleration}$$

density

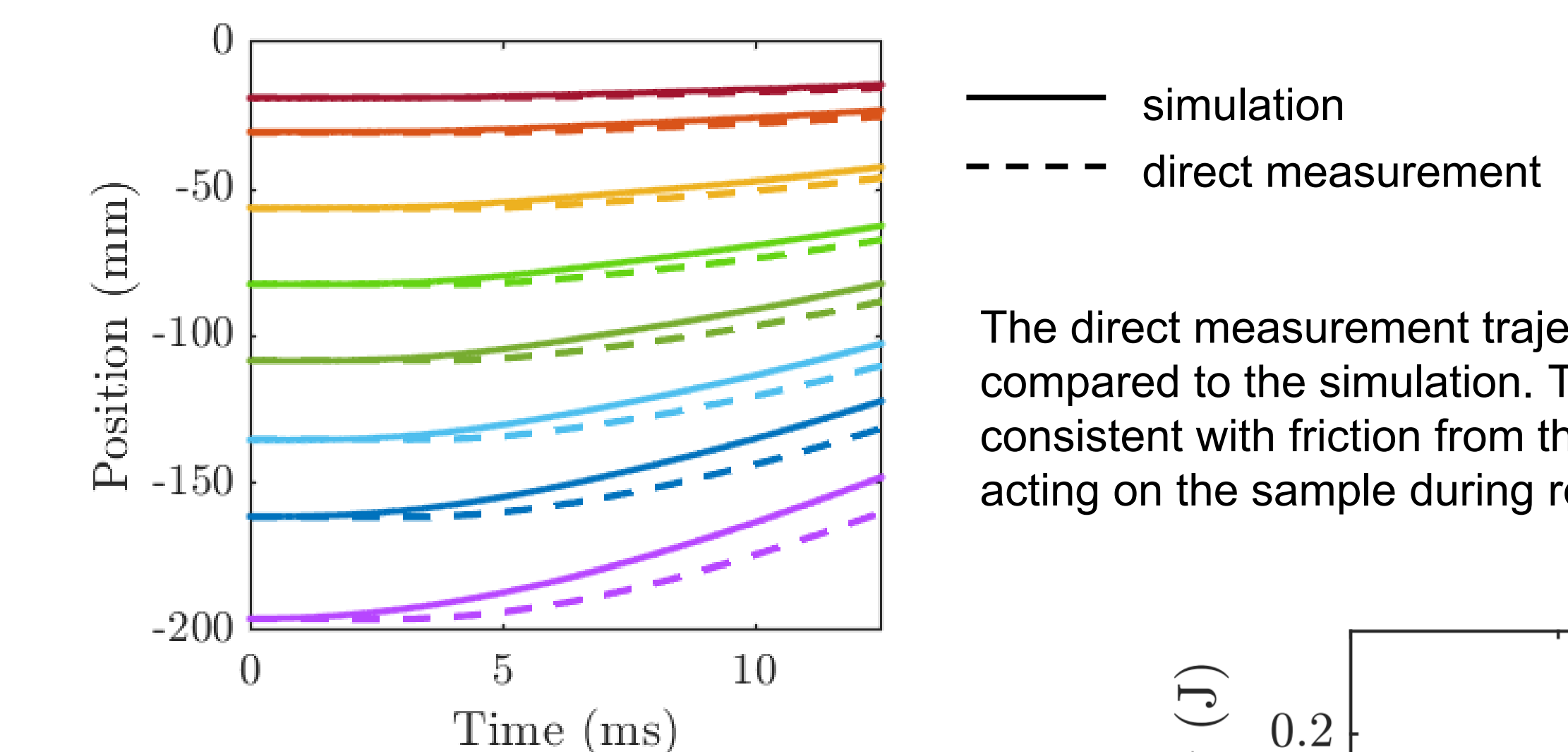
with the stress determined by the strain history $\epsilon(t)$ of the material at each point

$$\sigma(t) = \sigma_{\text{elastic}}(\epsilon_0) + \int_0^t \frac{E(t - \tau)}{E_{\infty}} \frac{d\sigma_{\text{elastic}}}{d\epsilon} \frac{d\epsilon(t')}{dt'} dt'$$



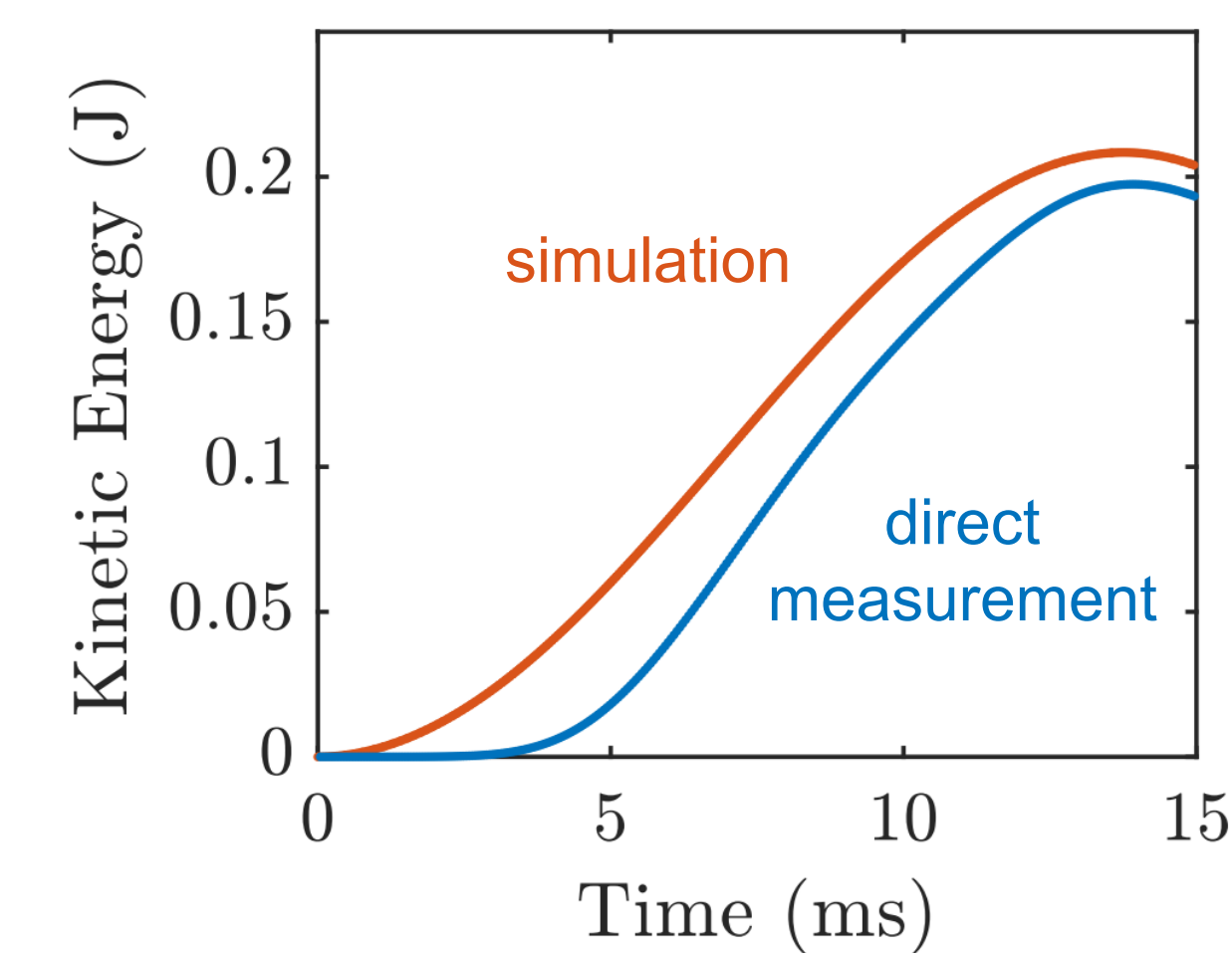
DISCUSSION

COMPARISON BETWEEN DIRECT MEASUREMENT AND SIMULATION



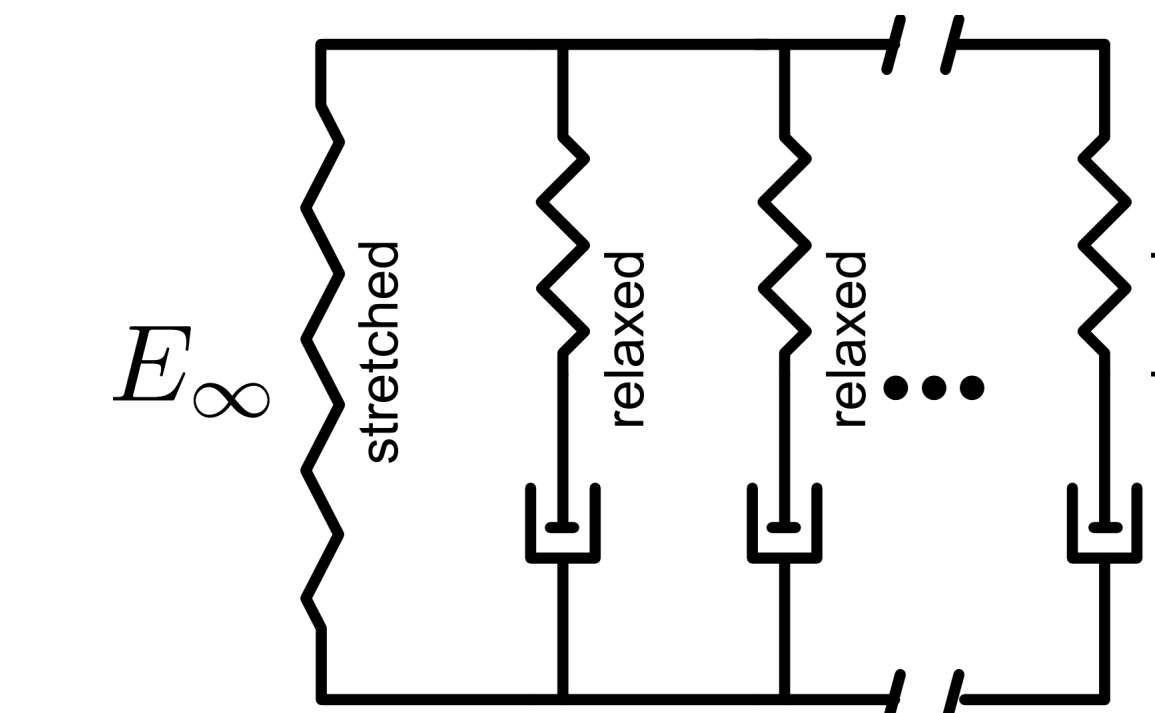
The direct measurement trajectories are delayed compared to the simulation. This discrepancy is consistent with friction from the pneumatic clamp acting on the sample during release.

The energy efficiency (resilience) of the material is lower during recoil (slow loading/fast unloading) than when the material is stretched and unloaded at a slow rate. This result is consistent with previous work (Tsai et al., 2024) that measured resilience at controlled rates.

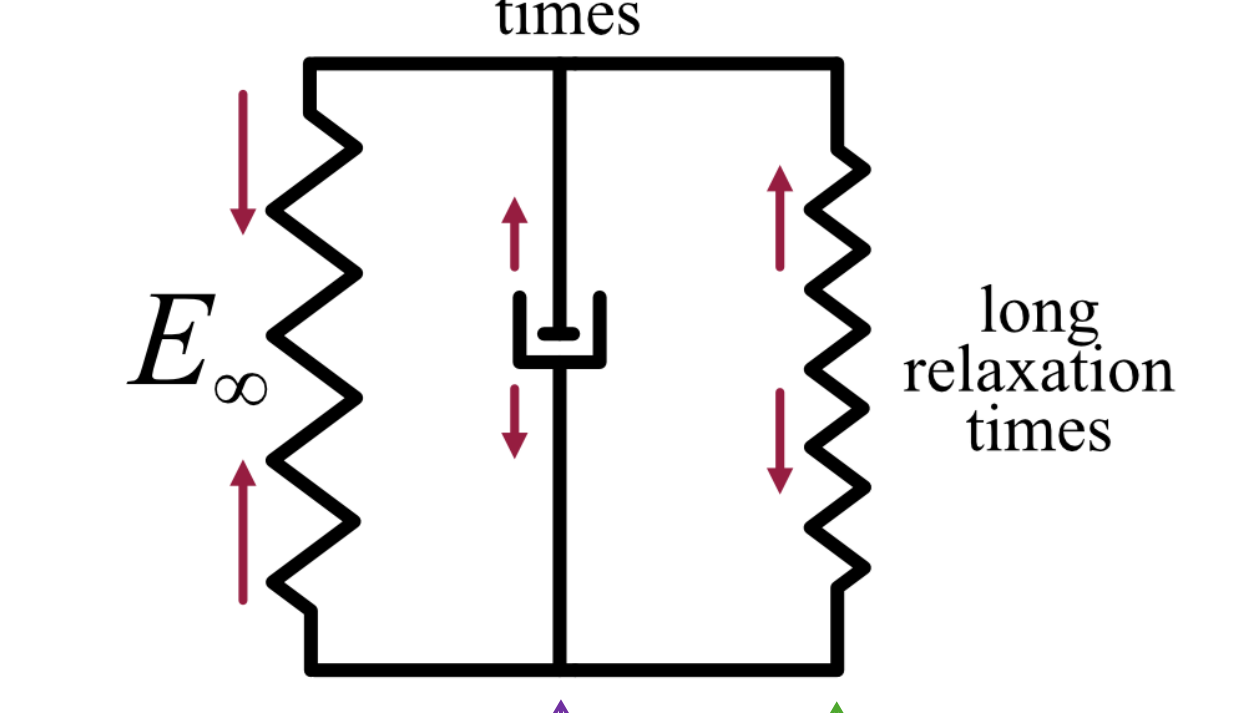


SIMULATION INSIGHTS

Before recoil



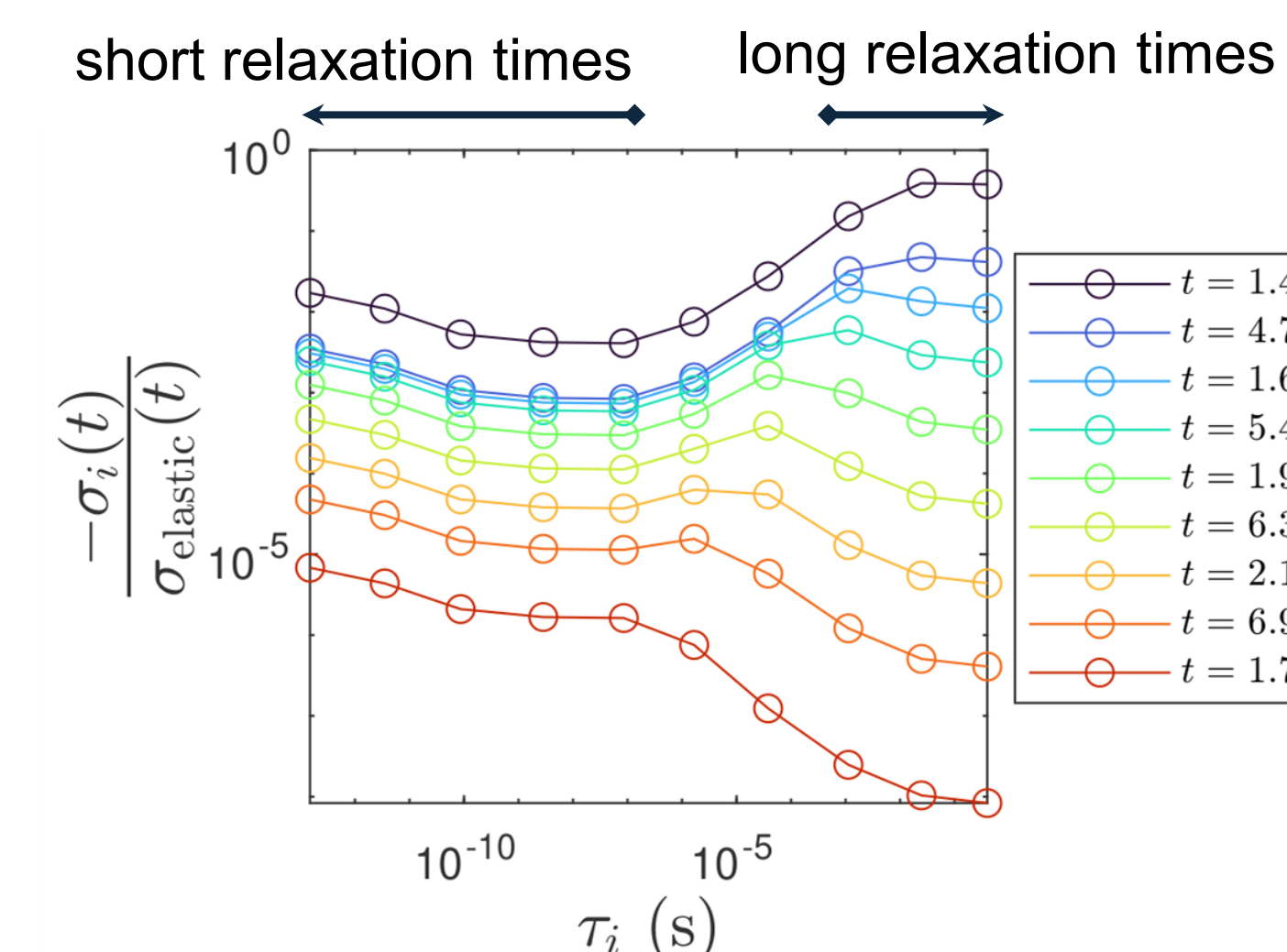
During recoil



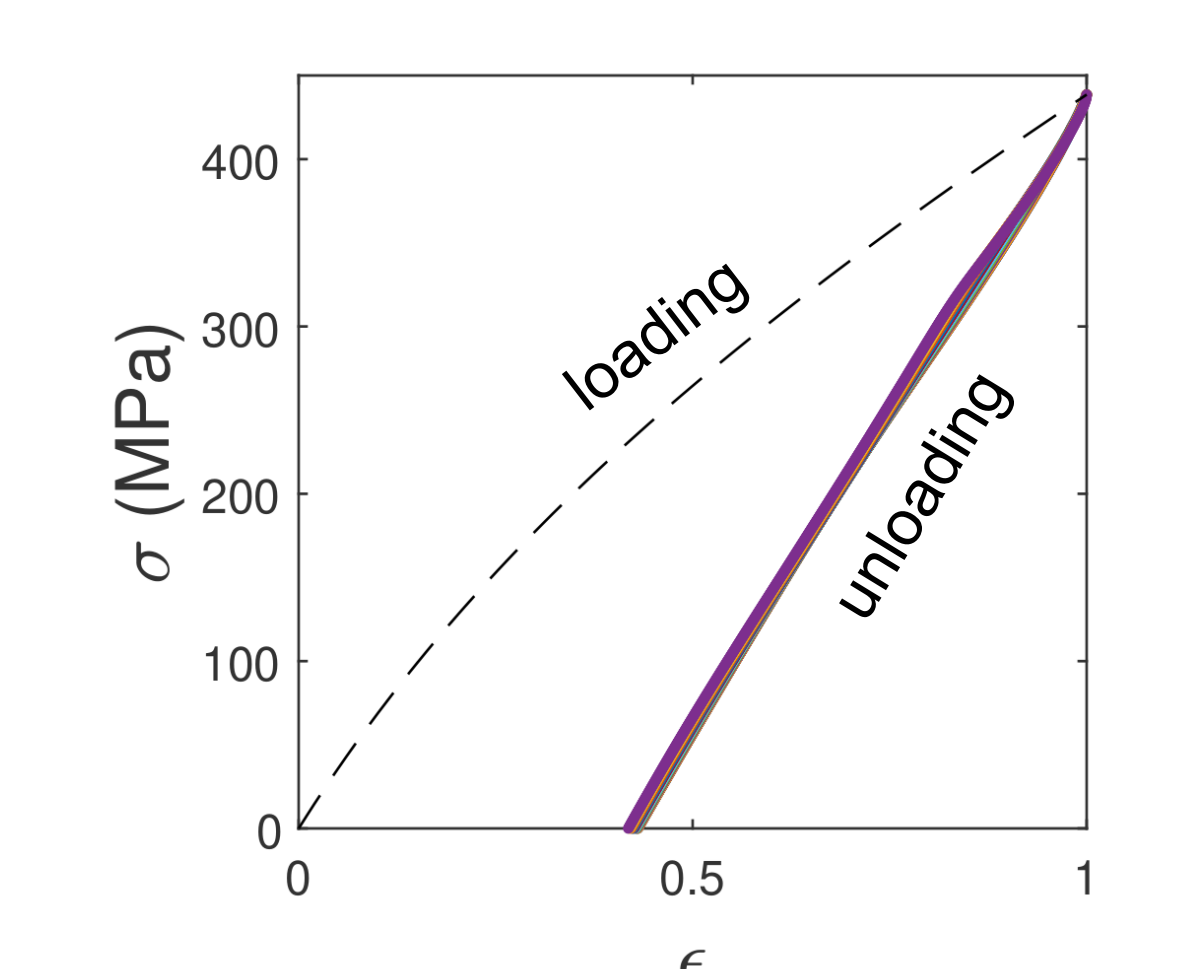
Short relaxation times act like dampers during the recoil and resist the recoil proportional to the strain rate, which increases with time.

Long relaxation times act like compressive springs that start in equilibrium just before recoil and provide an elastic resistance as they get compressed during the recoil.

Stress at end of band during recoil



Stress-strain behavior of each point



REFERENCES

Ilton Mark, Cox S.M., Egelmeers Thjis, P. Sutton Gregory, Patek S.N., and Crosby Alfred J. "The effect of size-scale on the kinematics of elastic energy release" Royal Society of Chemistry 2019

Tsai Lucien, Navarro Paco, Wu Siqi, Levinson Taylor, Mendoza Elizabeth, Janneke Schwaner M., Daley Monica A., Azizi Emanuel and Ilton Mark 2024 "Viscoelastic materials are most energy efficient when loaded and unloaded at equal rates" J. R. Soc. Interface.2120230527 <http://doi.org/10.1098/rsif.2023.0527>

Challita, Elia J., et al. "Slingshot spiders build tensed, underdamped webs for ultrafast launches and speedy halts." Journal of Comparative Physiology A 207.2 (2021): 205-217.